

$$5.0 \times 10^{-7} \text{ m} = \lambda$$

- 19 D** Electric potential energy $U = qV$

Since charge of electron is $-ve$, and potential at X is $+ve$ while that at Y is $-ve$,

$$\Rightarrow U_Y > U_X$$

Magnitude of electric force $F = qE$. Since the E field strength at Y $>$ E field strength at X

$$\Rightarrow F_Y > F_X$$

- 20 D** Break in two wires of cable $\rightarrow$ voltmeter reading $= 0$ when connected to X or Y